

1. When an entry in the LDIF has a multi value attribute, values are written:

- A. All values on a single line separated by whitespace
- B. One line for all values separated by double whitespaces
- C. One line for each value
- D. In a separate file LDIFMV(LDIF for multi value attributes)

2. While using JDBC a parametrized statement can be created with the method:

- A. prepareCall
- B. createStatement
- C. prepareStatement
- D. executeStatement

3. Which of the following statements is true?

- A. Primary keys uniquely identify a tuple
- B. Primary keys can have duplicate values
- C. Primary keys can have undefined values
- D. Primary keys uniquely identify a column

4. One privilege is:

- A. not right to execute a SQL command over an object from the schema
- B. a right to execute a SQL command over an object from the schema
- C. a right to execute a SQL command over an object from a different schema
- D. not a right to execute a SQL command over an object from a different schema

5. In a LDIF file for an entry, DN contains:

- A. none is correct
- B. only uid of the entry which is unique in the Directory Services
- C. only cn of the entry which is unique in the Directory Services
- D. the entire address of the entry which is unique in the Directory Services

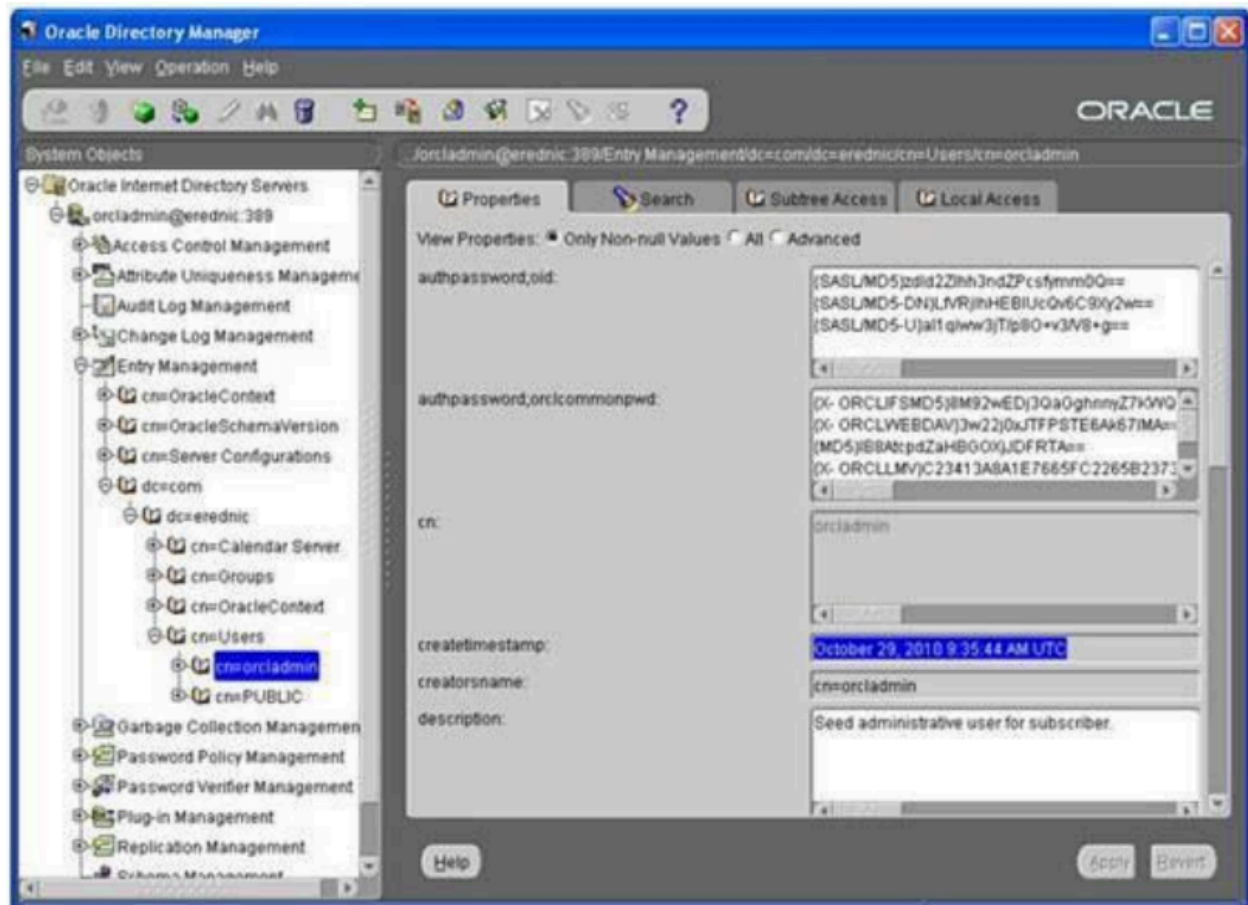
6. For what attributes the following reference is used "/":

- A. for clone attributes
- B. for binary attributes
- C. for files attributes
- D. for protoattributes

7. In JDBC a ResultSet object represents:

- A. The result of the execution of an SQL command
- B. A batch of SQL commands
- C. A connection
- D. An SQL command

8. The orcladmin user selected is:



- A. the superuser of the installed OID
- B. is a fresh user created in the rim
- C. none is correct
- D. is a simple user in the realm

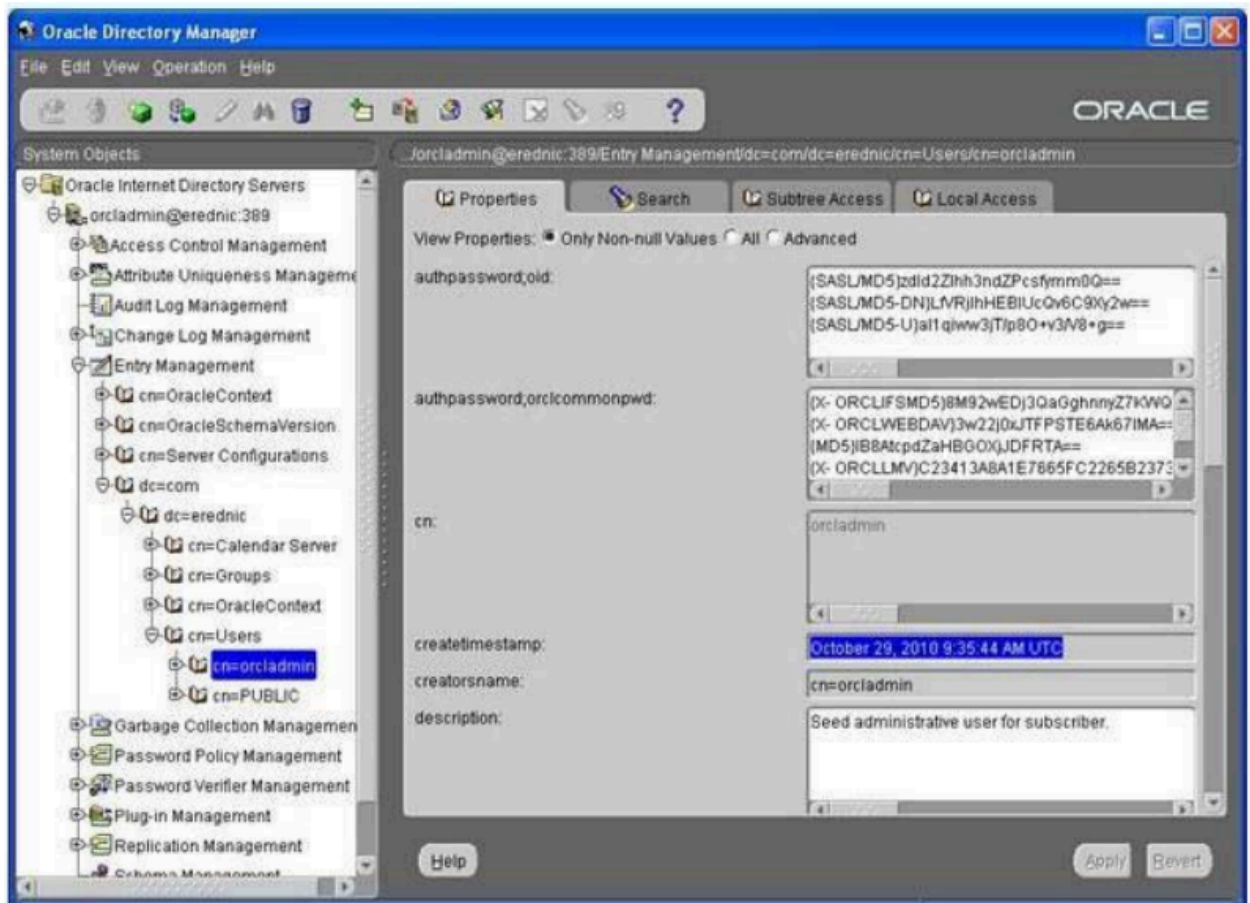
9. Which of the following is not a component of a table in relational model?

- A. Information about the physical allocation of the data
- B. The name of the attributes
- C. The name of the table
- D. Columns with values for each attribute

10. In a LDIF file, the lines in an entry continues on the next row using:

- A. ENTER
- B. TAB
- C. SHIFT
- D. CTRL

11. Which users are installed directly with OID in cn=Users:



- A. orcladmin and PUBLIC
- B. orcladmin, PUBLIC and PRIVATE, which is deleted in this installation
- C. none are correct
- D. orcladmin, oidadmin users

12. Which of the following is a DDL command:

- A. TRUNCATE
- B. DELETE
- C. INSERT
- D. UPDATE

13. Given a table with the structure: students (student id, student name, course name) can anomalies appear on an update to student_name?

- A. Yes
- B. No

14. Which of the following statements is true?

- A. In FN2 there are no transitive dependencies
- B. In FN1 redundant fields are accepted
- C. In FN3 there are no partial dependencies
- D. In FN3 transitive dependencies are accepted

15. What does the following code sequence return in MYSQL `aes_decrypt(aes_encrypt(s,key),key)`:

- A. The character string "key"
- B. A character string different from "s" or "key"
- C. An error
- D. The character string "s"

16. In JDBC a `CallableStatement` object is used for:

- A. The execution of a stored procedure
- B. The execution of a parametrized command
- C. None of the above
- D. Establishing a connection to the database server

17. Which of the following is true in the relational model?

- A. An attribute is a column of a table
- B. An attribute is a table
- C. An attribute is a line of a table
- D. An attribute is a data type

18. Which command is correct:

- A. `sqlplus /as system`
- B. `sqlplus /as sysdbadmin`
- C. `sqlplus /as sysdmin`
- D. `sqlplus /as sysdba`

19. Which of the following commands returns all clients who hi bought at least a product?

- A. `select* from clients where id-any(select cust_id from orders)`
- B. `select* from clients having id=any(select cust_id from orders)`
- C. `select* from clients where id = select cust_id from orders`
- D. `select* from clients having id=(select cust_id from orders)`

20. Which of the following is true?

- A. In FN2 there are redundant fields
- B. In FN2 there are transitive dependencies
- C. In FN3 there are redundant fields
- D. In FN3 there are transitive dependencies

21. Any privilege given to `PUBLIC` become a privilege:

- A. to any user from the database
- B. to the dropped users
- C. to nobody
- D. only to sysdba and sysoper users

22. Which of the following is a DML command?

- A. `INSERT`
- B. `CREATE`
- C. `DROP TABLE`
- D. `ALTER TABLE`

23. When logging to SSO with orcladmin user:



Is this:

- A. none is correct
- B. the realm orcladmin user
- C. the superuser
- D. the clone superuser

24. LDAP(Lightweight directory access protocol) is:

- A. none is correct
- B. not a relational database
- C. a model of distributed database
- D. a lite relational database

25.

```
dn: uid-marioara.popescu, ou=informatica,  
ou=Angajati, dc=mydomain, dc=com  
cn: marioara popescu  
sn: popescu  
givenname: marioara  
objectclass: top  
objectclass: person  
objectclass: organizationalPerson  
objectclass: inetorgPerson  
ou: informatica
```

```
ou: specialiste  
1: Bucuresti  
uid: marioara.popescu  
telephonenumber: +021 8889999
```

Telephonenumber is:

- A. mandatory attribute
 - B. none is correct
 - C. protoattribute
 - D. optional attribute
-
-

26. The definition of a table in the relational model does not contain:

- A. The name of the table
- B. The name of the attributes
- C. The way the data will actually be stored

27. Which of the following commands can have their effects reversed by rolling back a transaction:

- A. delete
- B. truncate
- C. drop

28. `CREATE USER 'andrei'@'localhost' identified by 'ism';` allows connections from:

- A. only a remote computer
- B. only localhost
- C. any computer

29. The current host has the ip 10.2.3.4. After the following commands:

```
GRANT select, insert, update on examples.* TO 'andrei'@'localhost';  
REVOKE insert, update on examples.* TO 'andrei'@'10.2.3.4';  
FLUSH PRIVILEGES;
```

The user can:

- A. Select data from the “examples” table
- B. Insert data into the “examples” table
- C. Drop the examples table

30. What does the following command `SELECT NAME FROM PROJECTS;` return?

- A. The names of the projects in the ascending order
- B. The names of the projects in descending order
- C. The names of the projects in the order they were inserted in

31. Which of the following SQL functions is not an aggregate function:

- A. COUNT
- B. AVG
- C. TRIM

32. What does the following command `Select category, avg(budget) from projects having budger>1000;` return:

- A. All the project categories with an average budger over 1k
- B. All the project categories
- C. Although it runs, the command contains a logical ERR

33. A resultSet object contains:

- A. a connection
- B. an SQL command
- C. the result of the execution of an SQL command

34. When using key generated by the database, they can be obtained through the _____ object:

- A. Statement
- B. Driver
- C. Connection

35. THE jdbc driver for MySQL cannot(CAN) be loaded through:

- A. The command line
- B. Loading the driver class in the class loader
- C. Registering a Driver instance with the DriverManager
- D. All of the above

36. A user can search for books stored in the DB through a web interface. The actual query should be implemented as a:

- A. Prepared Statement
- B. Statement
- C. CallableStatement

37. A table in FN1 does not contain:

- A. Transitive dependencies
- B. Null values
- C. Repeating groups

38. In the sequence `cStmt = conn.prepareCall("call ismtest(?)");` The “?” symbol can represent:

- A. Only an IN parameter
- B. Only an OUT parameter
- C. Both of them

39. Considering that “rsmd” is a resultSetMetaData object, what does the following statement return:

```
String catalog = rsmd.getCatalogName(1);
```

- A. the table name
- B. the database name
- C. the schema name

40. What does the integer array produced by the following statement contain:

```
int[] results = stmt.executeBatch();
```

- A. return codes for all executed operations
- B. the number of affected rows for each operation
- C. -1 for an operation if it did not succeed

41. Given the following code sequence, what will happen if the file cannot be opened:

```
try {
    conn = DriverManager.getConnection(url, user, password);
    BufferedReader reader = new BufferedReader(new
    FileReader("/path/to/file.csv"));
    // read from file and write to database
} catch (Exception ex) {
    Logger.getLogger(Main.class.getName()).log(Level.SEVERE, null, ex);
} finally {
    if (conn != null)
        conn.close();
}
```

- A. An exception will be thrown, but the connection to the database will be created and remain open
- B. An exception will be thrown, but the connection to the database will be created and be closed
- C. The connection will never be created

42. A callable statement object is used for:

- A. The execution of a stored procedure
- B. The execution of a DB function
- C. The execution of a DB trigger

43. Which of the following is true in the relational model?

- A. Rows can have different numbers of values
- B. Attribute values for each row can have different data types
- C. Rows can contain undefined values

44. Primary keys:

- A. Uniquely identify a tuple
- B. Can have undefined values
- C. Can have duplicate values

45. What does the column encpass contain after an insert, given the following trigger:

```
Create trigger upd_enc before insert on ism_passwords for each row set
new.encpass = encpassword(new.encpass);
```

- A. the values contained in encpass
- B. the values returned by the function encpasssword
- C. an insert would raise an error

46. Given the following code:

```
ResultSet rs = stmt.executeQuery("Select count(*) from employees");
```

Which is not a way to retrieve the total number of records from the result set:

- A. rs.getInt(1);
- B. rs.getInt(0)
- C. rs.getInt("Count(*)")

47. A table in fn2 does not contain:

- A. transitive dependencies
- B. functional dependencies of a subset of the key
- C. functional dependencies of any field

48. A table in FN2 is also in FN:

- A. 1
- B. 3
- C. 3.5

49. A table in FN3 does not contain:

- A. Functional dependencies of any field
- B. Transitive dependencies
- C. Multivalued dependencies

50. Which is the HIGHEST FN the table created below conforms to:

```
CREATE TABLE IF NOT EXISTS courses(  
Project_id int(11) NOT NULL,  
Project_name varchar(40) not NULL,  
Leader_name varchar(40) not NULL,  
Leader_address varchar(40) not null)
```

- A. 1
- B. 2
- C. 3

51. What does the following sequence return:

```
aes_decrypt(aes_encrypt(entity,key1),key2) ;
```

- A. The original content of entity
- B. The content of entity encrypted with key2
- C. Something else

52. Which command returns all the employees in all the projects as well as the project names without redundant information (an employee works on a single project):

- A. **Select** employees.employee_name,project.project_name **from** employees **JOIN** projects **ON** employees.project_id = project.project_id;
- B. **Select** employees.employee_name,project.project_name **From** employees **FULL JOIN** projects **ON** employees.project_id=project.project_id;
- C. **S Select** * **from** employees **JOIN** projects **ON** employees.project_id=project.project_id;

53. Which of the following commands returns all the students with an average grade over 8:

- A. **Select** students.student_name,AVG(grades.grade) **from** students **join** grades **ON** students.student_id = grades.student_id;
 - B. **Select** students.student_name, AVG(grades.grade) **from** students **join** grades **on** students.student_id= grades.student_id **GROUP BY** students.student_name **HAVING** AVG(grades.grade)>8;
 - C. **Select** students.student_name, **AVG**(grades.grade) **form** students **join** grades **on** students.student_id=grades.student_id **where** **AVG**(grades.grade)>8;
-
-

54. Which of the following is not a valid JDBC url:

- A. jdbc:mysql://nowhere.net:3306/acme
- B. jdbc://nowhere.net:3306/acme
- C. jdbc:odbc:x

55. The JDBC driver for MySql can be loaded through:

- A. the command line
- B. loading the class in the class loader
- C. registering a Driver instance with the DriverManager
- D. all of the above

57. Given the following code sequence:

```
PreparedStatement stmt = null;
stmt = conn.prepareStatement("UPDATE employees SET type = ? WHERE idemployees = ?");
stmt.setInt(2, 3);
stmt.executeUpdate();
```

What will happen?

- A. an exception will be thrown
- B. a warning will be thrown
- C. the "employees" table will be updated

58. In case of multiple attributes, how will they be displayed in LDIF?

- A. One line for all values separated by double whitespaces
- B. in a separate file LDIFMV(LDIF for multi value attributes)
- C. All values on a single line separated by whitespace
- D. One line for each value

59. What is a privilege?

- A. not a right to execute a SQL command over an object from a different schema
- B. a right to execute a SQL command over an object from the schema
- C. a right to execute a SQL command over an object from a different schema
- D. not a right to execute a SQL command over an object from the schema

60. What is a “dn” made of?

- A. only cn of the entry which is unique in the Directory Services
- B. the entire address of the entry which is unique in the Directory Services
- C. only uid of the entry which is unique in the Directory Services
- D. none is correct

61. For what is used “/” in LDIF?

- A. for files attributes
- B. for pro attributes
- C. for binary attributes
- D. for clone attributes

62. DIT inseamna:

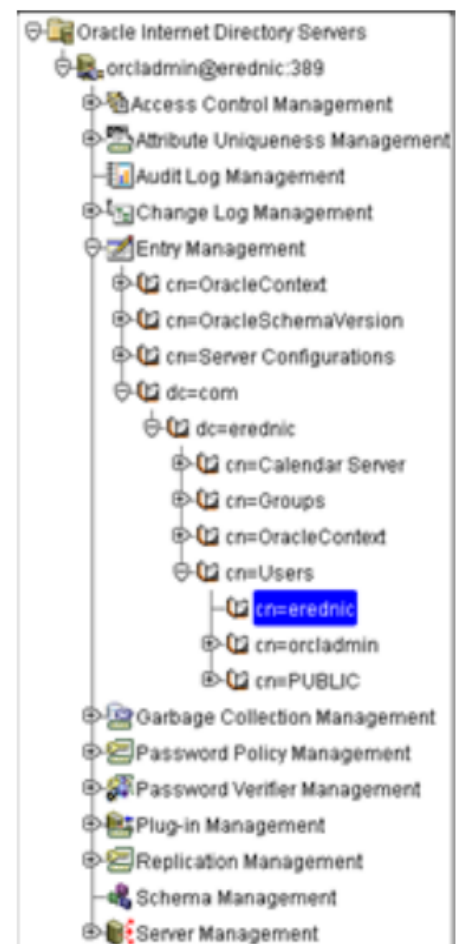
- A. Directory Information Terminal
- B. Direction Intersection Terminology
- C. Directory Informational Technique
- D. Directory Information Tree

63. In structura de mai sus: care este root-ul “radacina” OID-ului de mai sus:

- A. cn=users
- B. dc=com,dc=erednic
- C. dc=erednic,dc=com
- D. cn=users,dc=erednic,dc=com

64. Cati useri sunt in containerul cn=userr,dc=erednic,dc=com?

- A. 3
- B. 2
- C. 1, doar cel selectat
- D. nici unul



65. Comanda `ldabinentry` este folosita:

- A. pentru autentificare anonima
- B. pentru autentificarea cu un entry(user) folosit ca atribut
- C. nu este o comanda LDAP
- D. poate fi folosita pentru mai multe autentificari anonime

66. Fisierul `add_user.ldif`:

```
dn: cn=erednic,cn=users,dc=erednic,dc=com
changetype: add
cn: test1
sn: test1
userpassword:welcome1
mail: emanuil.rednic@gmail.com
objectclass: top
objectclass: person
objectclass: organizationalPerson
objectclass: inetOrgPerson
objectclass: orcluser
objectclass: orcluser2
```

Pentru ce tip de operatie LDAP este folosit?

- A. `Ldapadd`
- B. `Ldapadddn`
- C. `Ldappdn`
- D. `Ldapdnp`

67. Avem fisierul `add_user.ldif`:

```
dn: cn=erednic,cn=users,dc=erednic,dc=com
changetype: add
cn: test1
sn: test1
userpassword:welcome1
mail: emanuil.rednic@gmail.com
objectclass: top
```

Ce objectclass ar mai trebui adaugat?

- A. `Person`
- B. `inetOrgPerson`, `organisationalPerson`, `orcluser`
- C. `Person`, `orcluser2`
- D. `Person`, `inetOrgPerson`, `organisationalPerson`, `orcluser`, `orcluser2`

68. Autentificare cu un user specific se face folosind următoarea comanda:

- A. `Ldapbind -h host -p port -D "" -w userpassword`
- B. `Ldapbind -h host -p port -D "userdn" -w userpassword`
- C. `Ldapbind -h host -p port -G "" -w userpassword`
- D. `Ldapbind -h host -p port -G "userdn" -w userpassword`

69. Care comanda de mai jos returneaza primul user din containerul Users Din realmul: dc=erednic,dc=com?

- A. `ldapsearch -h host -p port -D "cn=orcladmin" -w password -z 1-b "cn=users"`
- B. `ldapsearch -h host -p port -D "cn=orcladmin" -w password -b "cn=users" -s one`
- C. `ldapsearch -h host -p port -D "cn=orcladmin" -w password -z 1-b "cn=users,dc=erednic,dc=com" -s one "objectclass=*" dn`
- D. `ldapsearch -h host -p port -D "cn=orcladmin" -w password -t 1-b "cn=users"`

70. Ldapcreate:

- A. E folosit pentru crearea unui nou entry in LDAP
 - B. E utilizat pentru crearea a uneia sau mai multe entry-uri in LDAP
 - C. Ajuta la stergerea unui entry sau mai multor
 - D. Nu este comanda LDAP
-

71. Care dintre cele de mai jos nu sunt privilegii SYSDBA?

- A. Startup
- B. Shutdown
- C. Hibernate
- D. Niciunul

72. Cine poate interoga v\$pwfile_users?

- A. scott b
- B. hr
- C. student
- D. sys

73. Ce este v\$pwfile_users?

- A. tabela virtuala
- B. partitie
- C. tabela
- D. niciun raspuns nu e correct

74. Care comanda este corecta?

- A. `select * from v$pwfiles_users;`
- B. `select count(*) from v$pwfile_users;`
- C. `select * from v$$pwfile_users;`
- D. `select * from v$pwfile_end_users;`

75. Ce reprezinta optiunea with admin option in comanda grant?

- A. Privilegiu ce poate fi transmis altui utilizator, de catre cel ce a primit respectivul privilegiu
- B. Privilegiu ce poate fi transmis altui utilizator, de catre toti utilizatorii, cu sau fara privilegii
- C. Nu este o optiune in comanda grant
- D. Niciun raspuns nu e corect

76. Ce reprezinta optiunea with nonadmin option in comanda grant?

- A. Privilegiu ce poate fi transmis altui utilizator, de catre cel ce a primit respectivul privilegiu
- B. Privilegiu ce poate fi transmis altui utilizator, de catre toti utilizatorii, cu sau fara privilegii
- C. Nu este o optiune in comanda grant
- D. Niciun raspuns nu e corect

77. Care comanda este corecta?

- A. **REVOKE DELETE ON** orders **FROM** hr;
- B. **REVOKE DROP ANY TABLES FROM** hr;
- C. **REVOKE DROP ON ANY TABLES FROM** hr;
- D. **REVOKE NOTHING FROM** hr;

78. Care din privilegiile de mai jos nu se mai regaseste din Oracle 9i, in sus:10g, 11g?

- A. create table
- B. create view
- C. create session
- D. create procedure

79. Ce tabele virtuale poate sa fie folosita pentru vizualizarea privilegiilor?

- A. all_tab_privs_made si user_tab_privs_made;
- B. all_tab_privs_granted si user_tab_privs_granted;
- C. all_tab_privs_made si user_tab_privs_made;
- D. all_tabs_priv_made si user_tabs_priv_made;

80. Privilegiu Caba nu poate fi atribuit rolului:

- A. Niciun raspuns nu e corect
- B. PRIVATE
- C. EXTENDED
- D. ADVANCED

81. Rolul PUBLIC:

- A. Este creat de scriptul sql.bsg care se ruleaza cand o baza de date se creaza
- B. Este creat de scriptul sql.bsq care se ruleaza cand o baza de date se creaza
- C. Este creat de scriptul sql.dsq care se ruleaza cand o baza de date se creaza
- D. Nu exista rolul PUBLIC

82. La nivel fizic, organizarea ORACLE este:

- A. Database ->tablespace->segment->datablock
- B. Database ->tablespaces->segments->datablock
- C. Database ->tablespaces->segment->datablock
- D. Database ->tablespaces->segment->datablocks

83. Tablespaces-urile temporare:

- A. nu exista tablespaces-uri temporare
- B. n- au tempfiles si au datafiles
- C. n-au decat datafiles
- D. au tempfiles si nu datafiles

84. Se da partitia:

```
create table sales (year number(4),  
product varchar2(10),  
val number(10,2))  
partition by range (year)  
partition p1 values less than (1992) tablespace u1,  
partition p2 values less than (1993) tablespace u2,  
partition p3 values less than (1994) tablespace u3,  
partition p4 values less than (1995) tablespace u4,  
partition p5 values less than (MAXVALUE) tablespace u5
```

Care afirmatie e corecta?

- A. Partitia nu e bine definita
- B. Partitia nu are definita MINVALUE
- C. Nu se pot introduce date in tabela partitionata mai mari decat MAXVALUE., unde MAXVALUE e max(coloana)
- D. Partitia nu are definit MAXVALUE corect

85. Care este relatia intre cluster si partition, la nivel fizic?

- A. nu exista nicio relatie
- B. de incluziune
- C. de excluziune
- D. niciun raspuns nu e corect

86. EUS vine de la:

- A. Enterprise Users Security
- B. Enterprise User Security
- C. Enterprises Unregistered Security
- D. Enterprise Unified Security

87. Securitatea bazei de date, este asigurata la nivelul integritatii datelor prin intermediul:

- A. DDL: grant/priv
- B. DML: grant/insert/delete
- C. DML grant/revoke
- D. DDL: grant/revoke

88. DES sau:

- A. Database encryption standard
- B. Data encryption support
- C. Data encryption standard
- D. Data encoded standard

89. Comanda idabindentry este folosita:

- A. Pentru autentificare anonima

- B. Pentru autentificare cu un entry(user) folosit ca atribut
- C. Nu este o comanda LDAP
- D. Poate fi folosita pentru mai multe autentificari anonime

90. Autentificare cu un user specific se face folosind urmatoarea comanda:

- A. Ldapbind -h host -p port -D "" -W userpassword
- B. Ldapbind -h host -p port -D "userdn" -w userpassword
- C. Ldapbind -h host -p port -G "" -W userpassword
- D. Ldapbind -h host -p port -G "userdn" -W userpassword

91. In JDBC, comenzile Sql simple se utilizeaza prin obiecte:

- A. Statement
- B. PreperedStatement
- C. CallableStatement
- D. ResaultSet

92. Pentru evitarea atacurilor prin acces direct la serverul de baze de date nu este utila:

- A. Criptarea conexiunii
- B. Criptarea la nivelul de linie
- C. Criptarea la nivel de tabela
- D. Criptarea la nivel de coloana

93. In JDBC un obiect CallableStatement se utilizeaza pentru:

- A. Executarea unei proceduri stocare
- B. Executarea unei comenzi parametrizabile
- C. Conectarea la o baza de date
- D. Niciun raspuns corect

94. Care comanda returneaza toti clientii care au cumparat cel putin un produs?

- A. `select * from clienti where id=any(select cust_id from comenzi);`
- B. `select * from clienti where id= select cust_id from comenzi;`
- C. `select * from clienti having id=(select cust_id from comenzi);`
- D. `select * from clienti having id=any(slect cust_id from comenzi);`

95. Care dintre afirmatii este adevarata?

- A. In FN1 pot exista campuri redudante
- B. In FN2 este nu exista dependente tranzitive
- C. In FN3 pot exista dependente tranzitive
- D. In FN3 nu exista dependente partiale

96. Care afirmatie este adevarata in modelul relational?

- A. Atributul este o tabele
- B. Atributul este o linie a unei tabele
- C. Atributul este o coloana a unei relatii
- D. Atributul este un tip de date

97. Este o comanda DML:

- A. INSERT

- B. CREATE
- C. ALTER TABLE
- D. DROP TABLE

98. Este posibil un atac prin Sql injection daca se utilizeaza:

- A. Obiect Statement
- B. Obiecte PreparedStatement
- C. Proceduri stocate
- D. Obiecte CallableStatement

99. Cheile primare:

- A. Pot avea valori nedefinite
- B. Identifica unic un tuplu
- C. Identifica unic o coloana
- D. Pot avea valori duplicate

100. Nu este o comanda MySql de management al rolurilor:

- A. CREATE USER
- B. REVOKE
- C. GRANT
- D. DELETE USER